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A study on the letter arrangement influence on eye typing efficiency

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Abstract

Various letter arrangements were analysed for the virtual Side keyboard developed during earlier studies. The key layout applied there, based on the users' opinions, could have been more convenient. Therefore, two additional solutions were introduced and verified. They were alphabetical and QWERTY-like letter layouts. Twenty participants were involved in the experiment, and the task was to type a sentence with their eyes. One sentence was used for all participants and all keyboards. Several metrics were studied: typing time, speed, and number of mistakes. The obtained results confirmed the previously collected opinions, favoring the alphabetical and QWERTY-like letter arrangements. It was especially visible for people without eye-tracking experience.

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Keywords: eye tracking; virtual keyboard; letter arrangement; typing speed and errors

1. Introduction

One of the main reasons for eye movement analysis is the possibility of its application for remote computer control. This way, users can manage cursor movement or select options on the computer screen through slight eye activities. Such a solution provides significant support for people who encounter difficulties in using traditional input devices, such as a keyboard and mouse. Eye-tracking is especially helpful for people with limited ability to move but still retain the ability to control eye movement, allowing them to communicate with the surrounding environment [12, 3]. It is not only more convenient for people with disabilities but also sometimes a more effective form of communication [13]. However, hygienic purposes can also be considered a field of virtual keyboard application, removing the need for touching screens on commonly used devices.

Controlling computers with eyes is often based on the analysis of fixations [24, 13] described by (x, y) coordinates of the user's gaze as well as their start time and duration. It creates different possibilities for the implementation of eye-computer interaction. Two approaches can be mentioned here: gaze-based control [21] and eye movement gestures [11, 2, 23]. Additionally, interaction by blinking eyes can also be listed [14].

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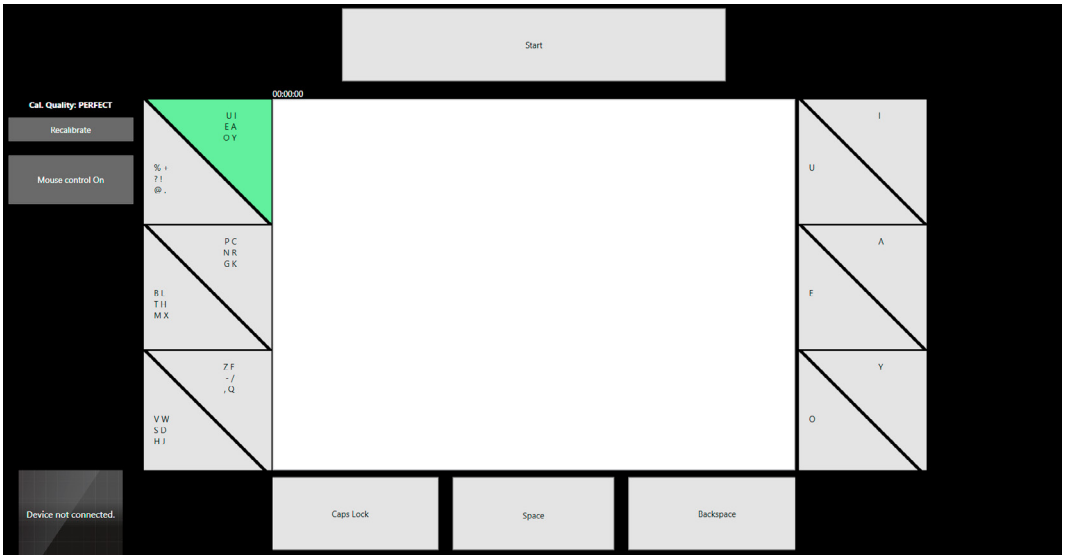


Fig. 1. The Side keyboard layout

The most natural selection method is to fix one's eyes on an on-screen object and gaze at it [9]. Three parameters for such an approach should be addressed: dwell time, object size, and content placement. A common issue when creating eye-controlled applications is the so-called Midas Touch Problem [13, 2] when users accidentally activate an action in a computer program because they unconsciously focus on its element. In such a situation, smaller elements or bigger gaps between them can be considered. The other approach is to adjust the gaze time on a component required to activate it. Both of these solutions have their drawbacks – too small elements may be unreadable, and extending their distance is impossible due to the available space in the interface. On the other hand, a more prolonged activation gaze extends the program operation time and negatively affects the user's concentration. Therefore, this time is usually between 500 and 1000 ms [10, 15, 22]. Some solutions use an adaptive time method. In such approaches, an algorithm adjusts the gaze duration according to the user's skills [16, 19, 8].

There are many possibilities for utilizing eye movements for man-machine interaction [4, 17, 6]. One of them is employing eye movement analysis to control a computer through virtual keyboards. Eye typing allows users to enter text and perform other tasks, such as changing an application setup [9] or manipulating calculators. It should be noted that not all eye-controlled keyboards are identical. Some may have higher precision or speed than others, and some do not require a complicated calibration procedure [1]. In the literature, several approaches were presented. They mainly differ in shape, layout of key buttons, and their dispersion [10, 20, 5, 18].

1.1. Motivation

The presented research is a continuation of studies regarding the development of a virtual keyboard [7] (Fig. 1). Two issues were considered when designing this *Side keyboard*: an intuitive choice of the desired letter and removing the necessity of using the BACK key when selecting the wrong group of letters. The key buttons were placed on the left and right sides of the screen. Those on the right are shown after choosing a letter group on the left. The buttons on the right have only one letter. They are arranged in the same order as in the group on the left. The shape of the buttons is meant to optimize the space – to fit as many groups of letters as possible in the smallest possible plane while, at the same time, maintaining a large area for easy gazing.

Entering letters using the Side keyboard is realized in two steps. The first step is to select the group of characters on the left screen side, which triggers the opening of the right panel with letters placed on separate buttons. During the second phase, users point out the letter they are interested in. When a character is selected, it is entered into the text field, and the entire right-side panel is hidden to indicate that the letter or symbol has been entered. The keyboard

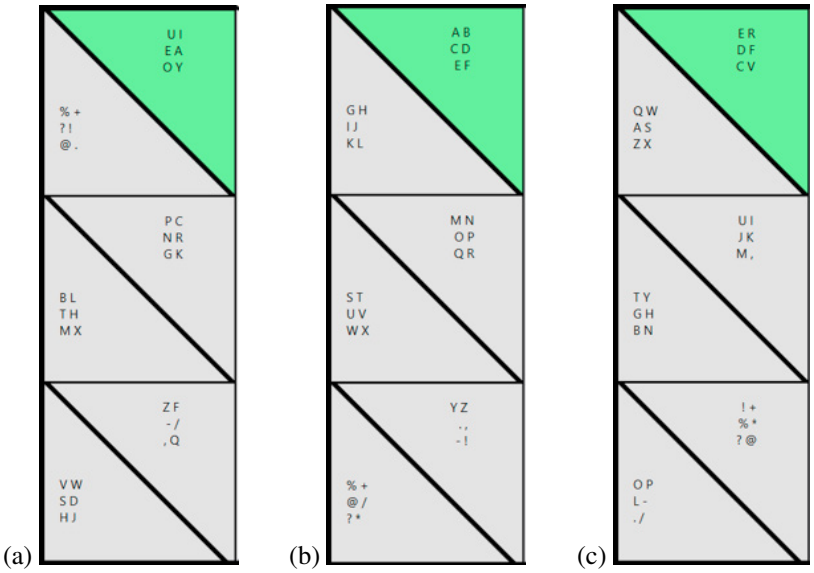


Fig. 2. Letter arrangements – a) default, b) alphabetical, c) QWERTY-like.

also has a graphical help when selecting buttons. When the gaze moves over a given triangle, it is highlighted in light green. After selecting this button by staring at it for a set duration, the triangle turns dark green.

This keyboard was assessed and compared with other solutions utilizing a similar typing approach [7]. The users rated it well but reported that the key layout could be more intuitive. Therefore, this research was devoted to exploring two other letter arrangements.

2. Method

As mentioned, based on the experience of the earlier conducted experiments [7] and users' opinions, the main goal of the current research was to compare three different letter orders:

- **default layout** – letters have been divided into groups based on their frequency of use. Vowels and special characters were placed at the top and bottom of the keyboard, respectively. Frequently used consonants have been placed in its central area (Fig. 2a).
- **alphabetical layout** -– the characters are arranged alphabetically, and special characters are placed at the bottom (Fig. 2b).
- **QWERTY-like layout** – letters are grouped in a way resembling the QWERTY-like layout known for traditional keyboards. This arrangement is intended to make finding letters easier for people who frequently work at a computer (Fig. 2c).

Additionally, a large 'Start/Stop' button above the text field allows for measuring typing time. Moreover, components related to calibration and the connection status of the eye recording are placed on the left part of the window. Based on the keyboards' initial tests, when participants became familiar with keyboards, the gaze time for selecting the keyboard element was set to 0.7 seconds – smaller values caused the Midas touch, and larger values quickly tired the eyes and unnecessarily prolonged the keyboard use. The adaptive algorithm proposed in [8] was omitted in the research to give all participants the same conditions for the experiment.

2.1. Experiment

The experiment participants were 20 men, aged 20 to 27, with secondary or higher education. The previous studies revealed that age, sex, and spoken language do not influence typing characteristics, which are rather related to personal

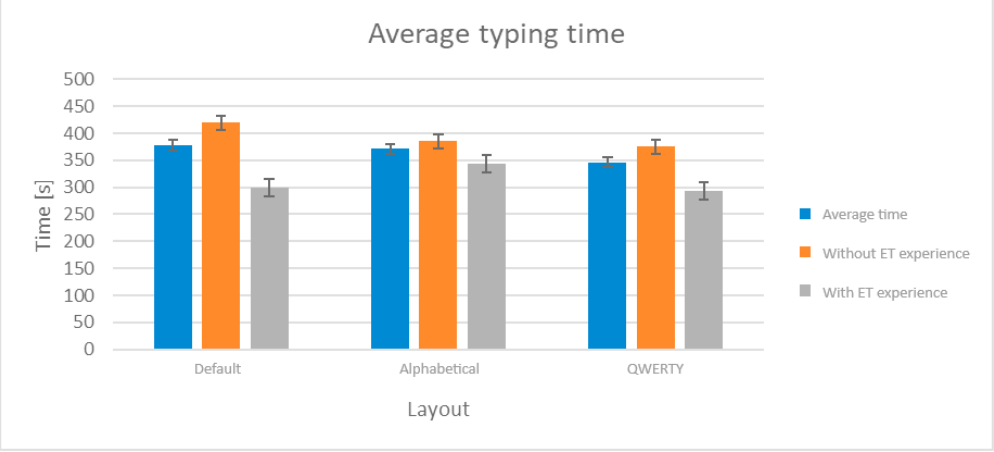


Fig. 3. Graph of the average typing time for three keyboards – blue for all participants, orange for participants without eye-tracking technology experience, and grey for subjects with such an experience.

traits [7, 8]. 8 users used glasses, and 7 out of 20 had already participated in eye-tracking experiments. 80% of participants use computers daily, so they were accustomed to the typical keyboard arrangement.

The subjects were first introduced to keyboard operation, which included a calibration process and, as part of becoming familiar with the application, writing the participant's name and surname on one randomly selected keyboard layout. The eye tracker was considered calibrated if the calibration quality obtained the best rating according to the device producer. In other cases, the calibration was repeated until successful. During the calibration, the subject followed a point presented on the screen in nine locations. After that, the subject's task was to write a sentence of 8 words and 75 characters with spaces and stop the time measurement. The sentence was defined in Polish: 'Zwariowany naukowiec odkrywa tajemnice w piwnicy opuszczonego laboratorium' (ang. A crazy scientist discovers secrets in the basement of an abandoned laboratory), and was entered with the usage of each keyboard. A second was subtracted from the final result because activating the "Stop" button took 0.7 seconds. Additional time was added to account for eye movement towards this button. The participants wrote the sentence using all three key arrangements in random order.

The study was conducted using an MSI GV62 laptop computer with a 15.6-inch screen and The Eye Tribe eye tracking device, which operated at a sampling rate of 30 Hz. The eye-tracker was located approximately 40 cm from the subjects' eyes. The tests were performed in a room with curtained windows and under the same lighting conditions.

During each trial, typing time, the number of letters written in one minute, and the number of errors made were collected. The person's age, education level, and experience in eye-tracking technology were also gathered. Any typos in the final sentence are considered errors – if the participant made a mistake while writing but corrected it, then it was not counted in the number of mistakes – but as a result, more time was needed to write the sentence.

2.2. Results

The best average typing time was observed for the QWERTY-like layout: 346.45 seconds. The graph in Fig. 3 shows, as expected, that the participants who had previously taken part in eye-tracking experiments entered the required text quicker than the others. In the case of the default system, the difference in average time between users without and with experience is as much as 120 seconds. The smaller outcomes dispersion was observed for the QWERTY-like letters order, while the lowest was recorded for the alphabetical arrangement.

The QWERTY-like keyboard was the most effective when typing speed was considered, with 13.56 letters per minute (Fig. 4). This result stems from the very good results obtained for the subjects experienced in eye tracking technology. They also performed well for the default layout; however, novices in controlling computers by gaze did the worst in that case, influencing the average outcome. For the alphabetical order, the difference in typing speed between experienced and inexperienced users is lower.

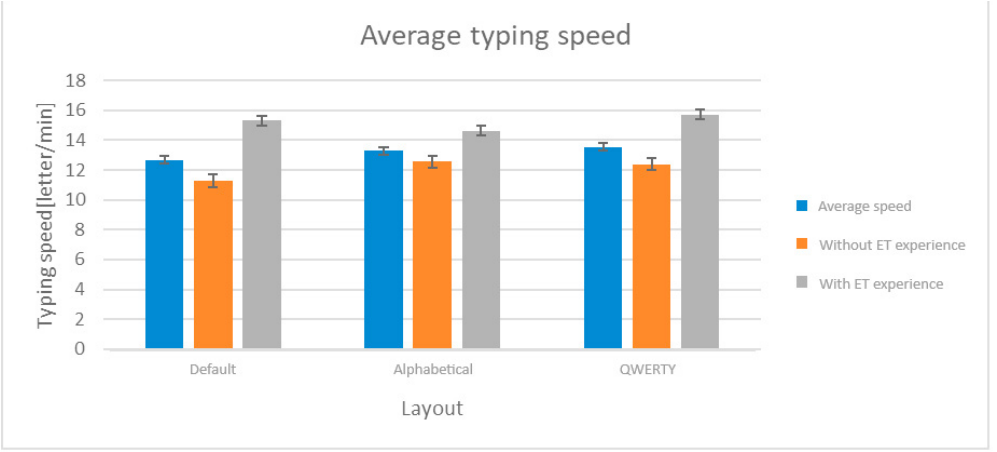


Fig. 4. Graph of the average typing speed for three keyboards – blue for all participants, orange for participants without eye-tracking technology experience, and grey for subjects with such an experience.

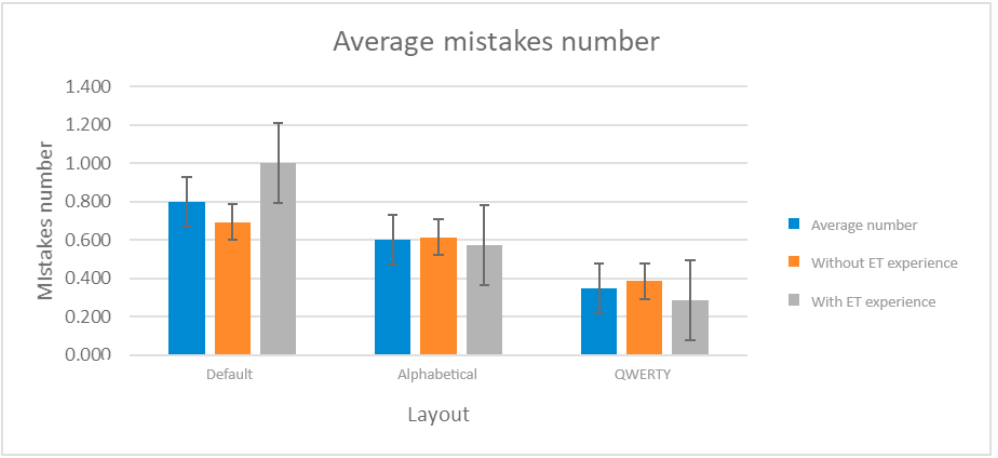


Fig. 5. Graph of the average mistakes number for three keyboards – blue for all participants, orange for participants without eye-tracking technology experience, and grey for subjects with such an experience.

The last analysed metric was the number of mistakes made during a trial. In this collation, the best result was again obtained for the QWERTY-like system – an average of 0.35 per participant. For comparison, 0.6 errors were made in alphabetical and 0.8 in default arrangements. Surprisingly, the experienced participants performed worse than the inexperienced for the default key order. These outcomes are presented in Fig. 5.

After comparing the average typing times for all keyboards and participants who made mistakes and those whose final sentences did not contain typos, it turned out that the subjects who wrote error-free sentences wrote them 10 seconds faster. Such a situation is slightly unexpected because error-free sentences also included sentences in which the subjects made a mistake but corrected it. Thus, this should have taken them more time than leaving a typo and continuing writing. It can be explained by the fact that some participants made no mistakes.

At the end of the experiment, after all trials, the participants were asked to assess each letter arrangement. They were expected to rate each keyboard from 1 to 5 and describe their general impression. Based on the provided opinions, it was revealed that despite better results achieved on the QWERTY-like keyboard, the subjects rated the alphabetical layout best. Fig. 6 shows a chart with the average ratings for particular arrangements. It can be seen that the participants' experience did not significantly impact the ratings. The default layout was rated the worst. The grades for individual users are presented in the Fig. 7.

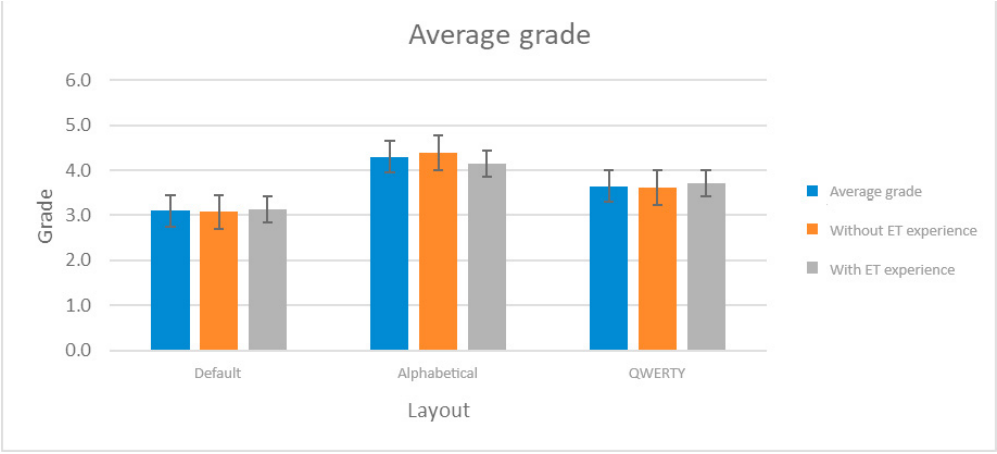


Fig. 6. Graph of the average user ratings for three keyboards – blue for all participants, orange for participants without eye-tracking technology experience, and grey for subjects with such an experience.

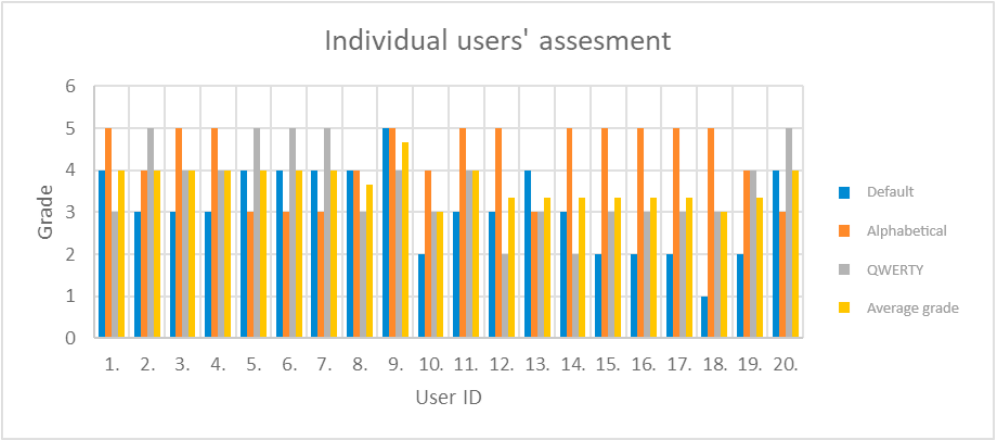


Fig. 7. Graph of the individual user grades for three keyboards – blue for default, orange for alphabetical, grey for QWERTY-like, and yellow representing average grade.

2.3. Statistical tests

The obtained results were verified regarding the statistical significance of their differences. At first, their normal distribution was checked using the Shapiro-Wilk test. It was conducted on outcomes for all subjects and groups with and without eye-tracking technology experience. A normality of all three keyboard layouts was revealed only for typing speed. In the case of typing time, the test failed for the default letter order (Time ALL) and for the alphabetical one (Time with ET experience), as shown in Table 1. All results for the number of errors made did not indicate a normal distribution. A normal distribution was observed for user ratings for the default layout.

Therefore, two methods were used to determine the statistical significance of the differences in the obtained results: ANOVA and Kruskala-Wallis tests. In all cases, the significance level was set to .05. The ANOVA test was performed on the results with a normal distribution, which included typing speed for all participants and participants divided into groups depending on eye tracking experience, typing time for inexperienced participants, and users' assessment for those with eye tracking experience. None of the tests revealed statistical significance (Table 2).

Subsequently, the Kruskala-Wallis test was used for the remaining metrics. In the case of typing time, the number of errors made, and experienced participants' assessment, no statistically significant differences were found. Such

Table 1. P-values for normality tests for typing time and user assessment.

Measure	DEFAULT	ALPHABETICAL	QWERTY-like
Time ALL	**.039**	.100	.578
Time without ET experience	.907	.810	.687
Time with ET experience	.132	**.030**	.479
User Assessment	.061	**<.0001**	**.011**

Table 2. Results for the ANOVA test

Measure	p-value	F-statistics	Test F
Typing speed	.704	0.353	3.159
Typing speed without ET experience	.445	0.829	3.259
Typing speed with ET experience	.842	0.174	3.555
Typing time without ET experience	.483	0.742	3.259
Users assessment with ET experience	.186	1.85	3.555

differences appeared in the case of user assessment for the whole group and those without experience. Based on Dunn's test, they occurred between the default and alphabetical arrangements, which can be seen in Table 3.

Table 3. Results for the Dunn's test

Group	Layout	Sum of ranks	Mean of ranks	Groups	
All participants	DEFAULT	434.500	21.725	A	
	ALPHABETICAL	803.000	40.200		B
	QWERTY-like	591.500	29.575	A	B
participants without ET experience	DEFAULT	183.000	14.077	A	
	ALPHABETICAL	350.000	26.923		B
	QWERTY-like	247.000	19.000	A	B

3. Discussion

When analysing the obtained results, it can be concluded that the QWERTY-like layout worked best. Not only was the lowest average typing time achieved (Fig. 3) and the highest typing speed (Fig. 4), but the fewest mistakes were also made (Fig. 5). However, despite the advantage of the QWERTY-like system, the alphabetical one received the best ratings from the participants. It achieved an average typing time of inexperienced people similar to the QWERTY-like keyboard and a relatively small difference in average times between participants with and without eye-tracking experience - about 40 seconds. When comparing the default and QWERTY-like letter orders, the average time improvement is 120 and 81 seconds, respectively.

The ANOVA tests showed that the differences in the data were not statistically significant. Therefore, it can be assumed that the QWERTY-like arrangement performed best due to participants' familiarity with the computer keyboard, but it was not statistically confirmed. Based on the Kruskal-Wallis tests, a statistically significant difference in user assessment was revealed for all participants and those without eye-tracking experience. It occurred between the default and alphabetical layouts, favoring the alphabetical one. Hence, it can be concluded that the alphabetical arrangement, unlike the default one, seems to be the most intuitive for beginners. In the case of users with experience, this difference is no longer significant.

4. Summary

The virtual Side keyboard analysed in the presented study was developed during the previous work, and it was positively assessed as a convenient tool according to the users' opinions [7]. However, the letter arrangement assumed when designing the keyboard turned out to be challenging. It was confirmed during the current exploration by the other group of users. The previous letter order was compared with alphabetical and QWERTY-like ones. Based on the obtained outcomes, both proved more effective and user-friendly than the default one. The alphabetical layout was especially appreciated by subjects inexperienced in eye-tracking technology. It can be expected that it will be an advantage for people who need to use a gaze-control keyboard but do not have prior skills for operating computers. Such tests are planned in further studies.

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